

Abhyuday Vaish

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science (CGPA: 9.01)

Chennai, India

July 2024 – May 2028

Delhi Public School

Class X: 95.6% — Class XII: 86.8%

Lucknow, India

2022 – 2024

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL (Postgres), HTML/CSS/JS, MATLAB

Frameworks: Flask, FastAPI, PyTorch, TensorFlow, LangChain, HuggingFace, Scikit-learn, OpenCV

Tools: Git, Docker, Cursor, Claude Code, Google Cloud Platform, Render, VS Code, Linux/Bash

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Linear Algebra, Probability & Statistics

CERTIFICATIONS

- **Deep Learning (NPTEL – IIT Madras):** Elite + Gold Certification (Score: 90%+).
- **Machine Learning Specialization (Stanford University):** Supervised, Unsupervised & Reinforcement Learning.
- **Generative AI for Software Development (DeepLearning.AI):** Applied GenAI patterns and LLM integration.

EXPERIENCE

Samsung PRISM Intern, Samsung R&D

March 2026 – Present | Remote

- Contributing to a quantum-inspired annealing project for multi-stage reasoning using small language models.

Research Intern, IIT BHU

March 2026 – Present | Remote

- Researching transcriptomics and deep learning-based drug repurposing for neurological and oncology applications.

AI/ML Development Intern, Daira EdTech

Aug 2025 – Oct 2025 | Remote

- Engineered an automated pipeline ingesting 100k+ UPSC PYQs, using OpenCV (adaptive thresholding, CLAHE) for high-accuracy text extraction from scanned documents.
- Architected a RAG/Generative AI system transforming static questions into dynamic assessments with auto-generated explanations, driving 40% user engagement growth.

Summer Research Intern, Vellore Institute of Technology

May 2025 – July 2025 | Chennai, India

- Pioneered opponent profiling by integrating Stockfish with Chess.com APIs, achieving 89.5% recall in identifying opening repertoire vulnerabilities.
- Fine-tuned a Gemma-2B LLM to generate automated counter-strategies, achieving 81.6% validation accuracy.
- Engineered a pipeline analyzing 2,500+ games to distill exploitable tactical patterns and vulnerabilities.

PROJECTS

Solar CME Detection System | Python, Flask, TensorFlow, PyTorch, Keras, PostgreSQL

Jan 2026 – Present

- Built a stacked LSTM pipeline forecasting Kp-index 1h ahead from 6h solar wind windows, benchmarked against PINN and Quantum architectures on 14,700+ samples (± 1 Kp tolerance).
- Engineered an ETL pipeline ingesting 20 months of Aditya-L1 ASPEX telemetry alongside real-time DSCOVR/NASA feeds into a PostgreSQL datastore.
- Built a Flask dashboard with 3-tier RBAC, real-time CME shock alerts via statistical anomaly detection, and scientist-restricted on-demand model retraining.

Aadhar Analytics Engine | Python, Pandas, Plotly

Jan 2026

- Implemented statistical Z-Score fraud detection ($Z > 3$) to flag daily volume spikes and distinguish valid enrolment camps from operational anomalies.
- Designed a resource optimization model to classify districts by saturation density, redirecting inventory effectively.
- Built a *Migration Radar* analyzing ~ 5 M transaction records to correlate address changes with fresh enrolments.

AI Academic Assistant | FastAPI, Next.js, Supabase, Playwright, Tesseract OCR, pgvector

June 2025 – Present

- Architected an AI academic assistant indexing 850+ PYQs across 170 courses with 5 context-aware chat modes for guided exam prep.
- Engineered RAG/OCR pipelines for automated note summarization and syllabus-aligned quizzes, saving students ~ 5 hours of manual revision per exam.
- Automated PYQ-trend-driven question paper generation across 8 syllabus modules to help students target high-weightage topics.
- Containerized all services with Docker and automated build-test-deploy workflows using GitHub Actions CI/CD, enabling seamless deployments to Render with zero manual intervention.

AWARDS & ACHIEVEMENTS

- **1st Place**, Industry-Academia Innovators Conclave (IAIC 2025): Research Poster Presentation Category.
- **Global Rank 754** in CodeChef Starters 224 (Max Rating 1450+).
- **Max Ratings:** LeetCode (1650+), Codeforces (1280+).

POSITIONS OF RESPONSIBILITY

Project Lead | AI/ML Team Member, Microsoft Innovations Club, VIT Chennai

Sep 2024 – Present | Chennai, India

- **Project Lead (Jan 2026 – Present):** Leading a 7-member team on the “AI Academic Assistant”; conducting code reviews, overseeing deployment, and mentoring juniors on GenAI.

- **AI/ML Member:** Organizing workshops and collaborating with the AI/ML team on Kaggle competitions and technical projects.

Data Science Member, Google Developer Groups (GDG), VIT Chennai

Sep 2025 – Present | Chennai, India

- Collaborating on data science projects and exploring new technologies with the community.